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Research proposal

Rock fabric, petrography, mineralogy, log analysis and core analysis of the Palaeocene Mey sandstone reservoir planned for CCS

ENVS608 Petroleum Reservoir Geoscience Research Project

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Non-expert summary

Human activities have increased the presence of carbon dioxide (CO₂) in the atmosphere. This emissions from the use of fossil fuels in electricity generation and industrialization affect the climate of the Earth. Options to store CO₂ instead of burning it into atmosphere have become important.

Carbon Capture and Storage (CCS) is a technology that can capture up to 90% of the carbon dioxide emissions and securely store it in depleted oil and gas fields or deep saline aquifer formations. One of these subsurface reservoirs that has been proposed as CO₂ storage site is the Paleocene Mey sandstone.

The Paleocene Mey sandstone is a sedimentological rock formation that was deposited in a specific environment. Particles were transported from the source area to sink area in the past. As it was deposited, it created the formation of sandstones that was buried approximately 60 million years ago in the present-day Moray Firth basin (UK North Sea).

Reservoir properties (porosity and permeability) of this sandstone dictate the quality of the storage. Pore space between particles (grains) is porosity and the rate in which fluids can flow in pore spaces is permeability. Size, sorting, orientation of these grains can affect the degree of porosity. The more porosity, the more carbon dioxide can be injected into the reservoir. Connectivity between pore spaces can be affected by diagenetic processes such as compaction and cementation, that reduce the quality of the rock and degree of permeability – the rate of fluid flow. Compaction is the process at which grains have been compacted closely and cementation is the process of filling the pore spaces with clay minerals. Additionally, rock strength of the potential storage unit can be affected by fluid pressure after compressed CO₂ injection into the rock.

The aim of this research is to determine the quality of the Paleocene Mey sandstone by the study of the mineralogy, rock fabrics and analysis of existing oilfield data made available by the company that produced oil and gas from the reservoir. The mineralogy will be derived using petrography (detailed description of the rock through microscopes) of the samples and give data about shape, size, orientation of grains, texture and grain contacts. Other instruments will be used to qualitatively define the particle size and bulk geochemistry to provide additional information about the rock such as the type of minerals that sandstone contains. The analysis of the existing information from the drilling wells such as core (samples of the drilled rocks) analysis data and wireline (measurements from the drilling site) data can help to combine and understand the variation within the rock formation of the interested sandstones.

Interpretation of the results will give more information about the quality of the Paleocene Mey sandstone and will assess whether this rock have a needed quality to be injected and store the CO₂ disposal. This research is a part of the big University project that is assessing the quality of different sandstones for planned CCS.

Background and Relevance

One of the main compounds that affect the climate of the Earth is Carbon dioxide (CO₂). Mitigating against climate change can be done by decreasing the emissions of the greenhouse gases. As a solution, subsurface aquifers such as hydrocarbon reservoirs, unminable coal-seams and saline aquifers have become as options to potential storage of huge volumes of carbon dioxide (Baines & Worden 2004).

In the UK part of the North Sea (Moray Firth Basin and Central North Sea), the Palaeocene Mey turbidite sandstones are proposed to be as potential reservoir to be used in Carbon Capture and Storage (CCS). These sand units are proven to bear hydrocarbon in pore spaces for several oil fields and have a trapping mechanism. The Mey Sandstone Member sits within the Lista formation and has a complicated depositional history (Kilham et al. 2014). Distribution and mineralogy of sand-prone intervals within mud-prone basinal formation is dependent on the sources of the turbidity and gravitational flows.

Porosity is the main factor to be considered because pore spaces dictate how much CO₂ can be stored. Core and log analyses revealed the range of porosity values. However, what controls porosity and how it varies away from the wells is not known. Permeability is an important quality as well due to control on rate of fluid flow during CO₂ injection. Additionally, the effect of fluid pressure on reservoir strength during the injection is also a cause of uncertainty. As a result, the following points are presented as aims of the project:

- 1) Quantitative assessment of the reservoir mineralogy (detrital and diagenetic mineralogy) and rock fabrics of samples derived from these sandstones and defining the shape, size, orientation of grains, and grain contacts nature using petrographic modal analysis and electron microscopy
- 2) Determination of the sediment grain size for the given core from the 2 wells
- 3) Determination of the core bulk geochemistry
- 4) Linkage of variable rock strength, quantitative mineralogy and fabric data
- 5) Relationship of derived data to existing core and log analysis data

This dissertation work is a part of a larger project at University of Liverpool that investigates the ability of different sandstones to CCS. University is a part of the ACORN consortium that is collaboration of universities and industrial partners to generate research and development for CCS realization. This project presents *scientific impact* not only in terms of CCS, but in terms of assessment of reservoir quality, mineralogy, petrography and geochemistry of turbidite type of the sandstones that has potential to be explored around the world and one of the example can be the recent discovery of Huntington field in overlying Sele formation (Edwards et al. 2017).

Work-plan / Methods

The focus of the project requires the use a combination of data and equipment from the Diagenesis Research and Rock Deformation laboratories.

Initial preparation to the project methodology is important. It will start with *trainings*, that is provided by the university laboratory technicians, to use Laser particle size analyser and Scanning electron microscope (SEM). After that, acquaintance with the principal techniques (tutorials) of optical mineralogy and petrography is required as well as acquaintance with provided thin sections and core – location of it, use, background and safety.

Data collection is going to start with the analysis of the previous data. Existing core and wireline data that was derived from the coring and core analysis (CCA) and wireline logs will be analysed. To better understand and use this information, format of the data should be converted in the uniform format (LAS/LIS). It can be done by using Schlumberger toolbox. Borehole seismic report is also presented in the files from the wells. Existed data such as Quantitative Evaluation of Minerals by Scanning (QEMSCAN), X-Ray diffraction (XRD) and geomechanical rock strength data are going to be analysed as well and upscale and link together with new data.

Core sections are derived from the two wells and have total length of approximately 50 m. According to the given data, it is mostly turbiditic sandstones that are interbedded with mud-prone formations. Description of the core will take two weeks and it will record the visual attributes of cored sections.

Core description is planned to be undertaken at the same time as portable X-Ray fluorescence (XRF) analysis of the core at all core analysis plug points. Samples will be collected adjacent to core plugs and analysed for grain size using Laser particle analysis. Use of XRF will also require training by skilled academic staff.

Thin section or light optics is next step of the work plan. It is going to occupy approximately one to two weeks. This analysis involves the use of optical microscope that is widely accessible in the university laboratories. Every thin section sample will be analysed using petrography methods such as point counting, modal analysis and rock fabric. SEM analysis is also part of petrographic analysis and will be done with the help of academic staff in CTL using the desktop Hitachi SEM.

Consequently, all given data that is related to the project will be transformed into integral/common database. This database will have information about wells, depths and correspond it for the all produced and existed data. Data analysis and interpretation is the next big step of the project workflow. All these data produced will be analysed using methods that will be discussed with project supervisors. After careful analysis and interpretation writing process is planning until the finishing the dissertation and final presentation.

Results

Types of data produced can be divided into following combinations:

- 1) Analysis of existing data (QEMSCAN, XRD, CCA, Wireline, rock strength) will result in variation diagram vs. depth and versus each other variable
- 2) Core description will result in sedimentary core log that will include general information by visual and hand lens assessment of the cored sections. It will be transformed into electronic core log using software such as EasyCore 1.3.12
- 3) Laser particle size analyser will give the quantitative sediment grain size distribution data from every core plug point (and probably petrographic sample and rock strength sample depths)
- 4) XRF will determine core bulk geochemistry from every core plug point and tell what kind of elements (and therefore minerals) that constitute the sediments
- 5) Petrography methods (light optics, SEM) will produce data that will determine the quantitative mineralogy (detrital and diagenetic) and help reveal the reservoir quality.

Interpretation

Analysis of the derived and existing data will lead to interpretation of the results. Detrital and diagenetic mineralogy will give information about what controls the rock quality (porosity and permeability). Degree of compaction, cementation, mineral type of diagenetic cement, location of the cement within pores, degree of dissolution of primary minerals, replacement this primary mineral to the secondary mineral. It is also important to interpret the texture of the primary sandstone and the proportion of interconnected porosity in terms of the assessment of permeability. All this information will help to expose the controlling factors on this important sandstone.

Additionally, this report may mention the rock strength because it is a function of detrital mineralogy and diagenesis (compaction, cementation). It can reveal the link between rock strength and mineralogy/fabrics in studied sediments.

References

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Resource justification

Data to be collected

Resource	Cost, £	Unit	Total cost, £
Core description	100 (in BGS)	50 m or 10 days	1000
Laser Particle size analyser	10	50 samples	500
XRF	100	50 samples	5000
Light optics (microscope)	20	7 days	140
SEM	200	3 days	600
Subtotal			7240

Table 1. Cost of the resources that will be used

Existing data

Resource	Cost, £	Unit	Total cost, £
QEMSCAN	n/a	n/a	n/a
XRD	100	n/a	n/a
Core analysis	n/a	n/a	n/a
Wireline analysis	n/a	n/a	n/a
Geomechanical rock strength data	n/a	n/a	n/a
Subtotal			6240

Table 2. Cost of the resources that have already been collected

Resources that is going to be used can be divided into two categories: to be collected and existing.

Existing data is given by the university for free. The cost of data collection is also will be covered by the university. However, the use of the samples, work itself and equipment used can be calculated in terms of the cost that is normally paid in the scientific projects (Table 1,2).

The project involves work in the laboratory facilities and use of the personal computer that are provided by the university for the academics for free. Related software also can be used such as the Schlumberger open-source application.

Gantt chart

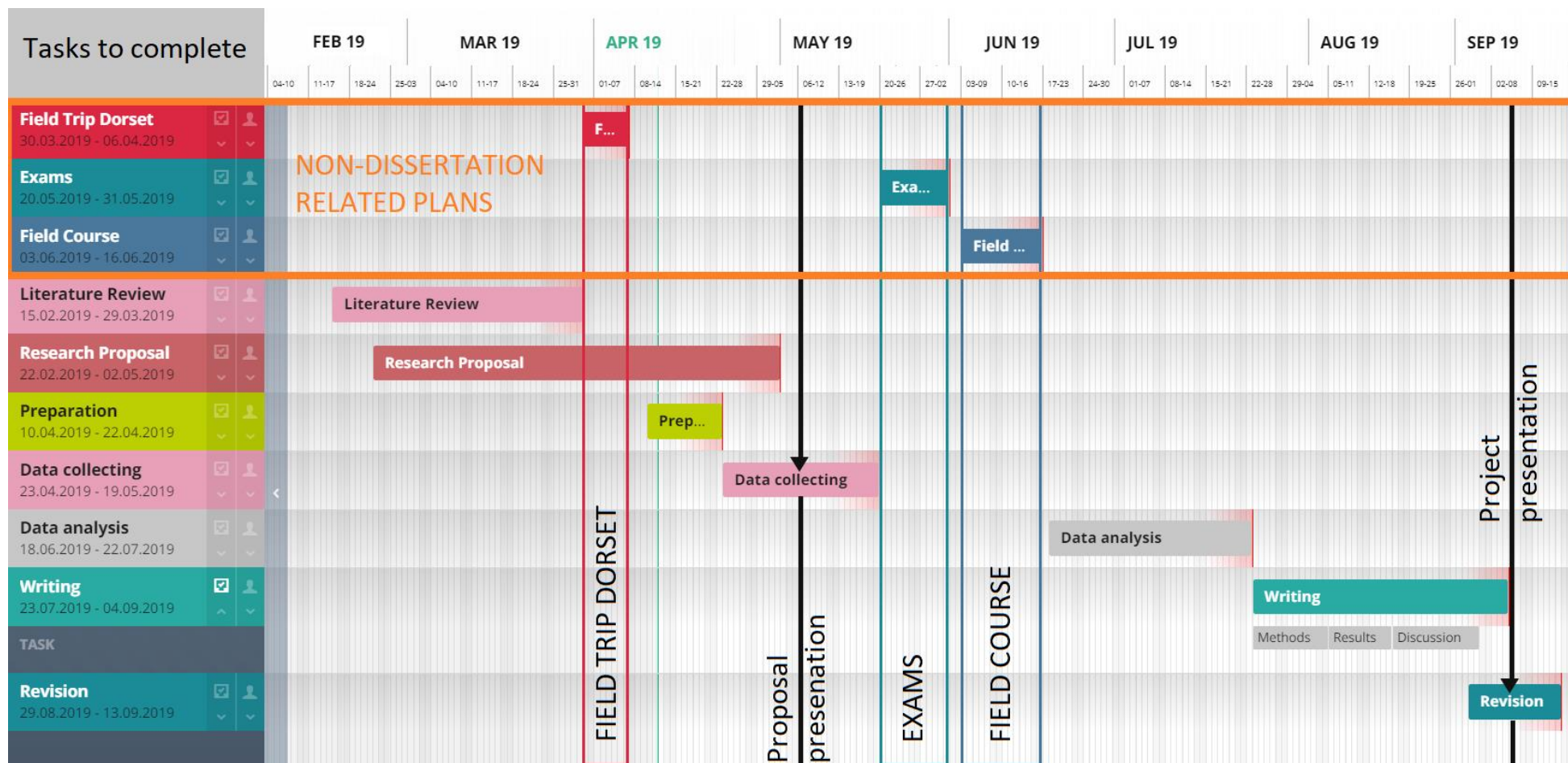


Figure 1. Gantt chart defining the work plan and timetable of the research.

Preparation includes trainings and acquaintance with the methods for the data collecting.

Data collecting will take 4 weeks and can be divided into three steps: 1. Core description and XRF - 1 week 2. Laser particle analysis - 1 week 3. Light optics and SEM - 1 week and additional week for the backup

